Debt but no taxation Example 3

Consider two firms which are identical except for their capital structure. Both will earn R300million in a boom and R100million in a recession. There is a 50% chance of each event. Firm U is financed entirely by equity. Its shares are valued at R1 000million. Firm L has issued R800million of risk free debt at an interest rate of 10%, and therefore R80million of income is paid out as interest. Assume there are no Taxes or any other market imperfections. Assume that investors can borrow and lend at a risk free rate of interest. Therefore:

Applying M-M:

(i) What is the value of L's shares

(ii) Suppose that you invest R40Million in U's shares. Is there an alternative investment in L that would give identical payoffs in a boom and recession? What is the expected payoff from such a strategy?

(iii) Now assume that you invest R40million in L's shares. Design an alternative strategy with identical payoffs.

(iv) What doe this say about capital structure?

Debt but no taxation Example 3 solution

(i) Because the firms are identical except for their capital structure, and there are no taxes, the total value of these firms must be the same. Thus, L's shares are worth R1000million minus R800million = R200million.

(ii) If you own R40million of U's ordinary shares , you will own 4% of the issued shares, hence, you will be entitled to:

0,04*R3000million = R12million if there is a boom and

0,04*R100million = R4million if there is a recession.

The equivalent investment is to purchasing 4% of L's issued shares which will cost 0,04*200million =R8million and to invest 0,04*800million = R32million in the risk free rate. Thus, the total invested for both is R40million. In a boom you will be entitled to: 0,10*32million + 0,04*(300million – R80million) = R12million and in a recession you will be entitled to 0,10*32million + 0,04*(100million –R80million) = R4million

Debt but no taxation Example 3 solution continued

(iii) If you own R40million of L's shares, you will own 20% of the issued shares , hence, you will be entitled to:
0,2*(R300milliom – R80million) = R44million if there is a boom and
0,2*(R100million – R80million) = R4million if there is a recession.
The equivalent investment is to purchase 20% of U's shares, which will cost 0,2*1000million = R200million but you only have R40million to invest, which means you need to borrow R160million at a risk free rate . Note that, the total Invested is the same at R40million.; Thus, in a boom you will be entitled to:
-0,1*160million + 0,2*R300million = R44million and in a recession you are entitled to:
-0,1*160million + 0,2*100million = R4million
(iv) Given the restrictive assumptions, this shows that the capital structure is Irrelevant. In all the situations, earnings were the same irrespective of the Capital structure.
However, in practice, the assumptions are not likely to apply

The Benefits and Costs of Using Debt

Effect of Debt Example 4

Vd

You are considering borrowing €1 000 000 at an
Ke
interest rate of 6 per cent for your pizza business. Your pizza business generates pretax cash flows of
Ve τ
€300 000 each year and pays taxes at a rate of 25 per
KA = Kos = 10%
cent. Cost of Assets , Kassets (which is equal to cost of ordinary shares (Kos) for the unlevered firm) is 10%.

What is the value of your firm without debt and how much would debt increase its value? What is WACC before and after restructuring?

VFirm = [€300 000 × (1 – 0.25)]/0.10 = €2 250 000

Value of tax shield = €1 000 000 × 0.25 = €250 000

Before: WACC $= K_{os} = 10\%$ before restructure

$V_F = V_e = 300k\,(1-25\%) \div 10\% = 2{,}25m$

After: $V_L = V_e + TS$

$= 2\,250\,000 + 1m \times 25\%$

$= 2{,}5m$

$V_e = 2{,}5 - 1m$

$= 1{,}5m$

After-tax CFs to shareholders

$= [300k - (1m \times 6\%)](1-t)$

$= 180k$

$K_{os} = \frac{180k}{1{,}5m} = 12\%$

$WACC = K_d\,W_d + K_e\,W_e$

$= 6\% \times \frac{1m}{2{,}5m} + 12\% \times \frac{1{,}5m}{2{,}5m}$

$= 9\%$

The Benefits and Costs of Using Debt

Effect of Debt Example 4 solution

WACC before financial restructuring = kos = 10%

After restructuring:

Value of the firm with debt = 2 250 000 + 250 000 = 2 500 000

Value of equity = €2 500 000 – €1 000 000

= €1 500 000

After-tax cash flows to shareholders

= [€300 000 – (€1 000 000 × 0.06)] × (1 –0.25)

= €180 000

kos = €180 000/€1 500 000 = 0.12, or 12 per cent

WACC = (€1 000 000/€2 500 000)(0.06)(1 – 0.25)

+ (€1 500 000/€2 500 000)(0.12)

= 0.09, or 9%

The Benefits and Costs of Using Debt

The perpetuity model assumes that

1. The firm will continue to be in business forever.

2. The firm will be able to realise the tax savings in the years in which the interest payments are made (the firm's EBIT will always be at least as great as the interest expense).

3. The firm's tax rate will remain constant.

Exh16.6: How Firm Value Changes

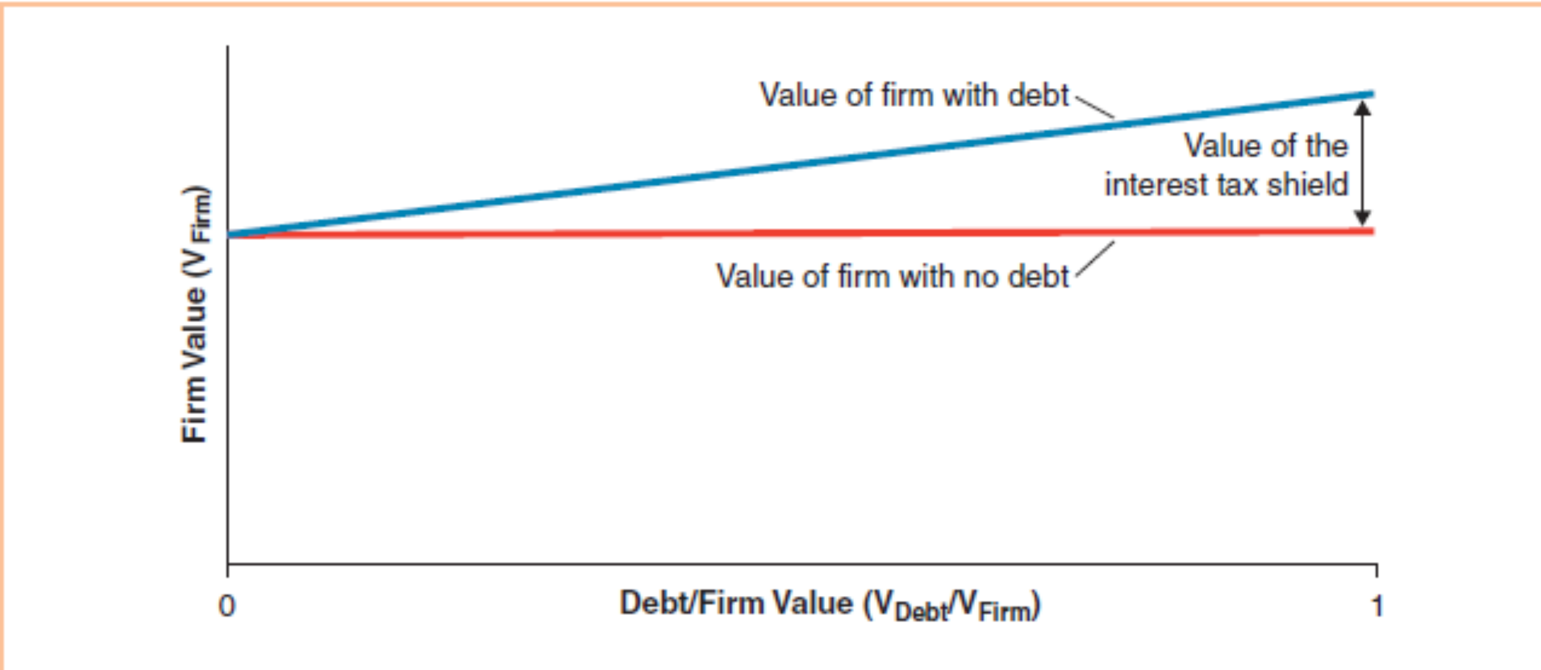

Exhibit 16.6: How Firm Value Changes with Leverage when Interest Payments are Tax Deductible and Dividends are Not The value of a firm increases with leverage when interest payments are tax-deductible and dividends are not and when the second and third M&M conditions apply; namely, there are no transaction costs and the real investment policy of the firm is not affected by its capital structure decisions.

Exh16.7: The Effect of Taxes on the Firm Value and WACC of Millennium Motors

EXHIBIT 16.7

THE EFFECT OF TAXES ON THE FIRM VALUE AND WACC OF MILLENNIUM MOTORS

Total debt	€ 0	€ 200	€ 400	€ 600	€ 800
Cost of debt	5.00%	5.00%	5.00%	5.00%	5.00%
EBIT	€100.00	€100.00	€100.00	€100.00	€100.00
Interest expense	—	10.00	20.00	30.00	40.00
Earnings before taxes	€100.00	€ 90.00	€ 80.00	€ 70.00	€ 60.00
Taxes (35%)	35.00	31.50	28.00	24.50	21.00
Net income	€ 65.00	€ 58.50	€ 52.00	€ 45.50	€ 39.00
Dividends	€ 65.00	€ 58.50	€ 52.00	€ 45.50	€ 39.00
Interest payments	—	10.00	20.00	30.00	40.00
Payments to securityholders	€ 65.00	€ 68.50	€ 72.00	€ 75.00	€ 79.00
Value of equity	€650.00	€520.00	€390.00	€260.00	€130.00
Cost of equity	10.00%	11.25%	13.33%	17.50%	30.00%
Firm value	€650.00	€720.00	€790.00	€860.00	€930.00
WACC	10.00%	9.03%	8.23%	7.56%	6.99%

The value of Millennium Motors increases and its WACC decreases with the amount of debt in the capital structure. The calculations assume that the cost of debt remains constant regardless of the amount of leverage and that the second and third M&M conditions apply.

Other Benefits of using debt

- Debt provides managers with incentives to focus on maximising the cash flows that the firm produces since interest and principal payments must be made when they are due.
  - Because managers must make these interest and principal payments or face the prospect of bankruptcy, not making the payments can destroy a manager's career.
- Debt can be used to limit the ability of bad managers to waste the shareholders' money on things such as fancy jet aircraft, plush offices and other negative-NPV projects that benefit the managers personally.

The Costs of Debt

- Financial managers limit the amount of debt in their firms' capital structures in part because there are costs that can become quite substantial at high levels of debt.
  - At low levels of debt, the benefits are greater than the costs and adding additional debt increases the overall value of the firm.
  - At some point, the costs begin to exceed the benefits and adding more debt financing destroys firm value.
  - Financial managers want to add debt just to the point at which the value of the firm is maximised.

Exh 16.8: Trade-Off Theory of Capital Structure

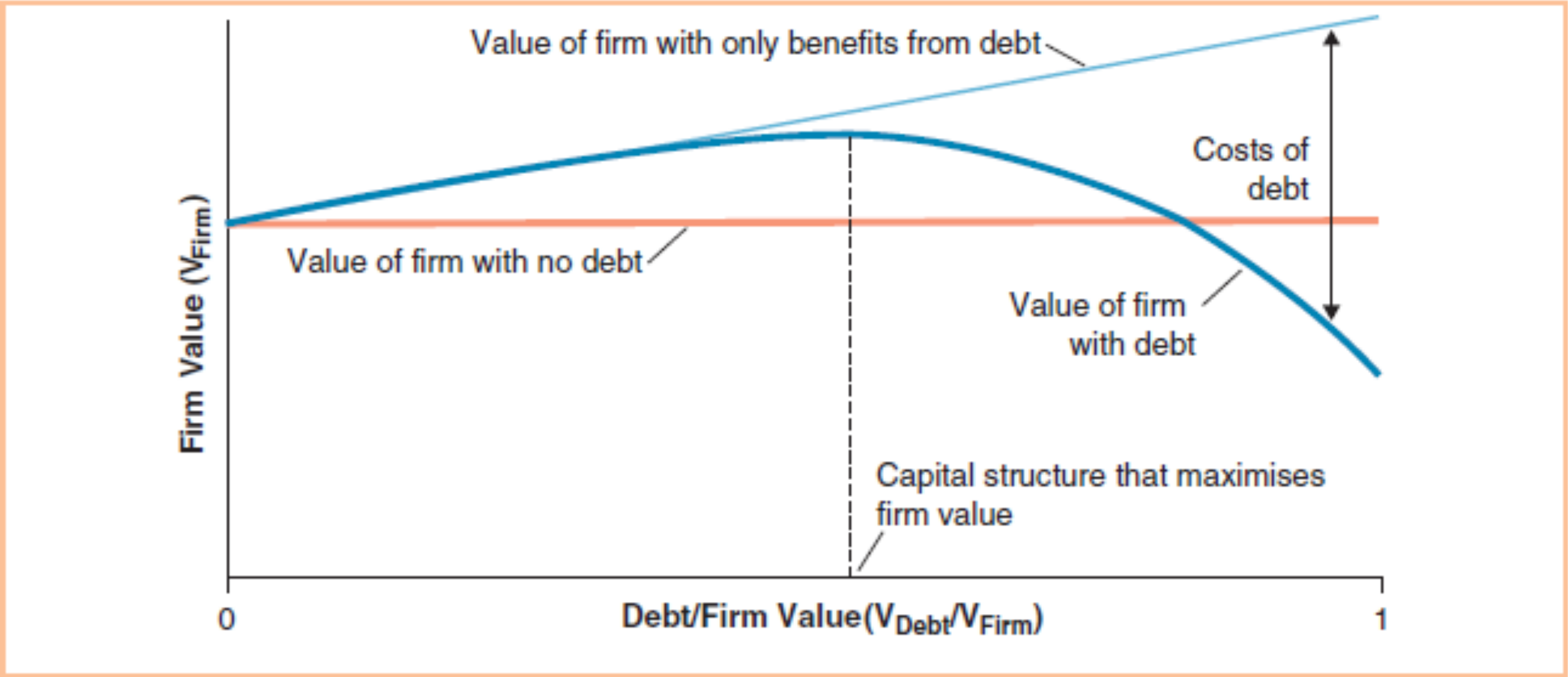

Exhibit 16.8: Trade-Off Theory of Capital Structure The benefits and costs of debt combine to affect firm value. For low levels of debt, adding more debt to a firm's capital structure increases firm value because the additional (marginal) benefits are greater than the additional (marginal) costs. However, at some point, which is the point at which the value of the firm is maximised, the costs of adding more debt begin to outweigh the benefits and the value of the firm decreases as more debt is added. The difference between the upward-sloping line and the curved line reflects the costs associated with debt.

The Costs of Debt

- Bankruptcy costs, also referred to as costs of financial distress, are costs associated with financial difficulties that a firm might get into because it uses too much debt financing.

- The term bankruptcy cost is used rather loosely in capital structure discussions to refer to costs incurred when a firm gets into financial distress.

- Firms can incur bankruptcy costs even if they never actually file for bankruptcy.

The Costs of Debt

- Direct Bankruptcy Costs are out-of-pocket costs that a firm incurs as a result of financial distress.
  - They include things such as fees paid to lawyers, accountants and consultants.
- Indirect Bankruptcy Costs are costs associated with changes in the behaviour of people who deal with a firm in financial distress.

The Costs of Debt

Some of the firm's potential customers might decide to purchase a competitor's products because of:

Concerns that the firm will not be able to honour its warranties.

Parts or service will not be available in the future.

➢ Some customers will demand a lower price to compensate them for these risks.

The Costs of Debt

- When suppliers learn that a firm is in financial distress, they will worry about not being paid; to protect against losses for future shipments, they often begin to require cash on delivery.

- Employees at a distressed firm will worry that their jobs or benefits are in danger and some start looking for new jobs.
- If the firm enters into the formal bankruptcy process, it incurs another indirect bankruptcy cost because a bankruptcy court must approve all investments made by the firm.

The Costs of Debt

- Agency costs result from conflicts of interest between principals and agents where one party, known as the principal, delegates its decision-making authority to another party, known as the agent.
  - The agent is expected to act in the interest of the principal but agents sometimes have interests that conflict with those of the principal.

The Costs of Debt

- Shareholder-Manager Agency Costs occur to the extent that if the incentives of the managers are not perfectly identical to those of the shareholders, managers will make some decisions that benefit themselves at the expense of the shareholders.
- Using debt financing provides managers with incentives to focus on maximising the cash flows that the firm produces and limits the ability of bad managers to waste the shareholders' money on negative-NPV projects.

The Costs of Debt

- These benefits amount to reductions in the agency costs associated with the principal-agent relationship between shareholders and managers.

- While the use of debt financing can reduce agency costs, it can also increase these costs by altering the behaviour of managers who have a high proportion of their wealth riding on the success of the firm, through their shareholdings, future income and reputations.

The Costs of Debt

- The use of debt increases the volatility of a firm's earnings and the probability that the firm will get into financial difficulty.

- Increased risk causes managers to make more conservative decisions.

- Shareholder-Lender Agency Costs can occur when investors lend money to a firm and delegate authority to the shareholders to decide how that money will be used.

The Costs of Debt

- Lenders expect that the shareholders, through the managers they appoint, will invest the money in a way that enables the firm to make all of the interest and principal payments that have been promised.

- However, shareholders may have incentives to use the money in ways that are not in the best interests of the lenders.

The Costs of Debt

- Lenders know that shareholders have incentives to distribute some or all of the funds that they borrow as dividends and so they protect themselves against this sort of behaviour by including provisions in the lending agreements that limit the ability of shareholders to pay dividends and conduct other behaviours.

- These protections are not foolproof.

The Costs of Debt

- One example of this behaviour is known as the asset substitution problem, where once a loan has been made to a firm, the shareholders have an incentive to switch from less risky assets to more risky assets, such as negative-NPV projects.

The Costs of Debt

- Another example behaviour is known as the underinvestment problem and it occurs in a financially distressed firm when the value that is created by investing in a positive-NPV project is likely to go to the lenders instead of the shareholders; therefore the firm forgoes financing and undertaking the project.